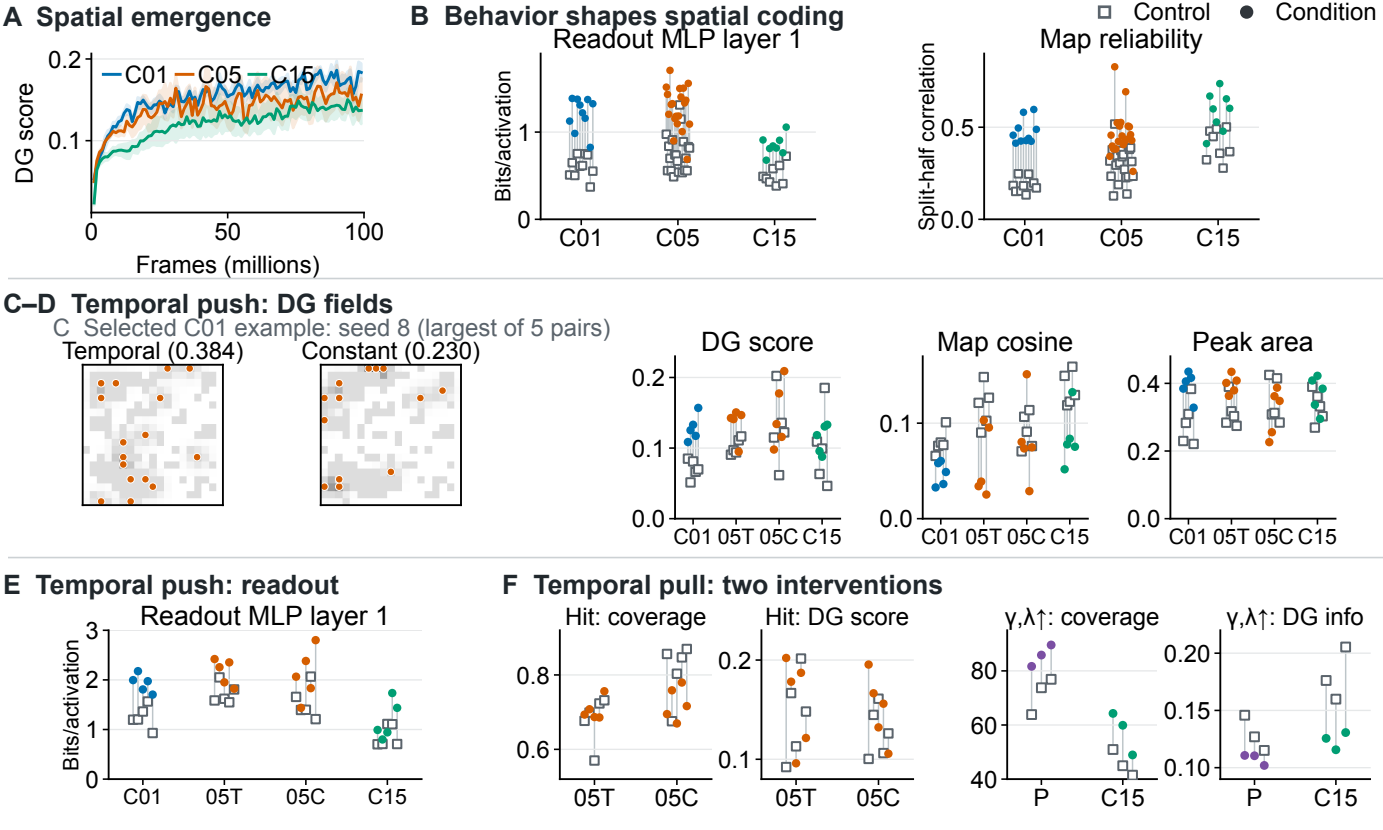

A Spatial emergence
DG score
0.2
0.1
C01
C05
C15
0
50
100
Frames (millions)
B Behavior shapes spatial coding
Control
Condition
Readout MLP layer 1
Bits/activation
1
0
C01
C05
C15
Map reliability
Split-half correlation
0.5
0.0
C01
C05
C15
C–D Temporal push: DG fields
C Selected C01 example: seed 8 (largest of 5 pairs)
Temporal (0.384)
Constant (0.230)
DG score
0.2
0.1
0.0
C01 05T 05C C15
Map cosine
0.1
0.0
C01 05T 05C C15
Peak area
0.4
0.2
0.0
C01 05T 05C C15
E Temporal push: readout
Readout MLP layer 1
Bits/activation
3
2
1
0
C01
05T
05C
C15
F Temporal pull: two interventions
Hit: coverage
0.8
0.6
05T
05C
Hit: DG score
0.2
0.1
05T
05C
γ,λ↑: coverage
80
60
40
P
C15
γ,λ↑: DG info
0.20
0.15
0.10
P
C15